ICT Corner

CRYSTAL FIELD THEORY

By using this tool you can understand the crystal field splitting of different metal ions in octahedral and tetrahedral ligand field and also calculate the Crystal Field Stabilisation Energy (CFSE) of a complex using crystal field theory.

Please go to the URL http://vlab.amrita.edu/index.php?sub=2&brch=193&sim=610&cnt=4
(or) Scan the QR code on the right side

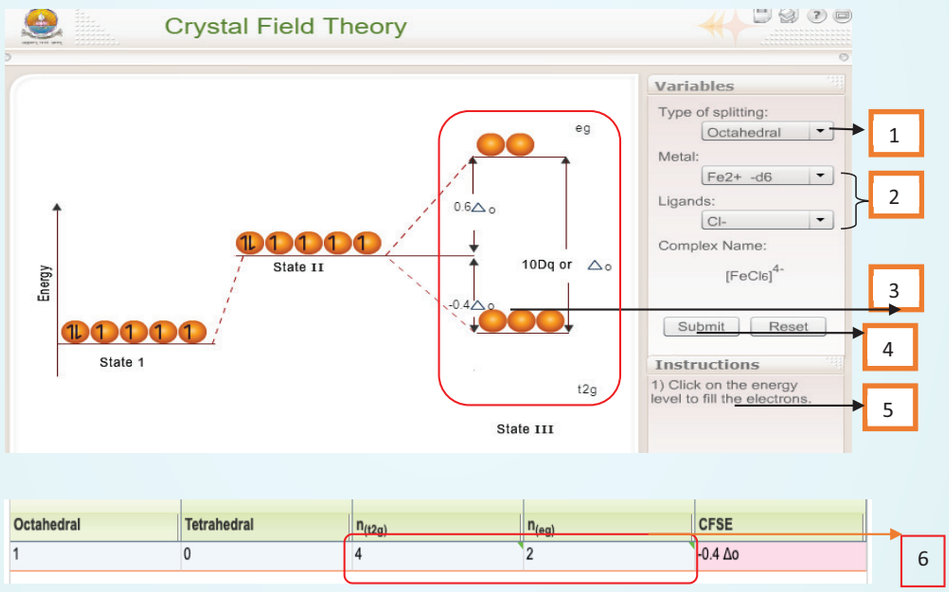
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Steps

- Open the browser and type the URL given (or) Scan the QR Code. You will see the webpage as shown in the figure.

  Note: *One time sign up is needed to access this webpage. Login using your username and password. Once logged in click the simulation tab.*

- You can select a suitable ligand field splitting using the drop down menu (box 1). Select a metal of interest and a ligand using the drop down menu (box 2). Now crystal field splitting for the selected complex appears on the screen.
- Apply crystal field theory to the selected complex and fill the d-electrons in the t_{2g} and eg orbitals by clicking each orbital. Click on the orbitals thrice to remove electrons. After completion, click submit button (box 4). Now you can check the correctness of the electron distribution. If wrong try again.
- Enter the number of electrons in the t_{2g} & e_g orbitlals in the work sheet at the bottom of the page (box 6), The calculated Crystal Field Stabilisation Energy (CFSE) will be displayed.



Octahedral	Tetrahedral	$n_{(t2g)}$	$n_{(eg)}$	CFSE
1	0	4	2	-0.4 Δo

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